

# KPIs to check and consequences on social features

High Level

# Social: goals

2 different directions:

1. Social = Ability to bring free new players
  - a. The key action is out-game (new player accepts to install the game)
  - b. The main goal is acquisition
2. Social = Ability to interact with so called friends in the game
  - a. The key actions are in-game
  - b. The main goal is usually retention, but monetization is an important goal too

# Different kinds of friends

- The split between in-game friends and out-game friends is not enough
- Indeed, even if the closest friends are out-game friends (real friends, family, etc.), some out-game friends can be strangers, especially FB friends
- Combined with in-game/out-game criterion, this scale can be used:
  - Strangers: they exist only in games
  - Acquaintances: the player know who they are, but doesn't have direct relations with them outside the game (coworkers from long time ago, friends' family members, ...)
  - Friends: the player has direct relations with them outside the game
  - (Close) Relations: close family or close friends. The player has direct relations with them outside the game, either frequent or intimate (or both)

Social acquisition

# KPIs, basics

- Main KPI is K-factor
  - We will focus on **direct K-factor** (the number of new players directly recruited) and not limit K-factor (the number of new players recruited by the original player, the player he/she has recruited, etc.). Limit K-factor has a time dimension which makes it hard to use
- Once K-factor known, additional indicators are
  - **Recruited New Players (RNP) quality**: how do they perform compared to acquired players
  - We would like to have RNP answer time, needed number of invitations, install/invite rate but we don't have access to that
  - Instead we have Inviting Player data: time between last invitation and current RNP, ratio between number of time invitation was clicked and current number of RNP
  - We also have **Inviting Player status** when inviting: where is he/she inviting from (what was the prompt)? When in his/her game progression did they invite (true or late game for example)?

# Basics first consequences, I

- If K-factor is very low, there is most probably a huge obstacle to find before creating any new feature. If K-factor is already very high, it will be difficult to have a real impact
- RNP quality will have an impact on the budget that can be used for the features. Or the shift of goal from “higher K-factor” to “better recruitment” (most probably linked to social retention: what do I do with my friends once they are in the game?)

# Basics first consequences, II

- If invitation is effective (low answer time, low number of invitations needed), all the energy should be put on increasing the conversion invitation opportunity VS real invitation
  - A/B testing the rewards (not only quantity but also type) should be one of the first moves
- If invitation is not effective, it's still important to make the player click "invite" more often, but it's not enough. Finding an attractive message for potential players (who don't know the game yet) becomes more important.
  - We need to rely on elements everybody knows
  - Example from Midcore games: "one SSR fighter guaranteed" or "200 free 3\* summoning"
  - Giving a higher value to the invitation (specific opportunity: if you install from this link, you will have more) can help too

# Basic first consequences, III

- Player status has 2 sides: weaknesses and motivation
- Of course, first move will be to work on weaknesses: if the player doesn't invite during FTUE, FTUE invitation must become more attractive for example
- But then, focusing on "where it works" should be the next priority, not to change or optimize (not immediately fix what is not broken), but to understand the player motivations
- These motivations can then either be reused elsewhere, in the "weaker" situations, or help create and experiment new motivations
- **We need to understand as much "why they invite" as "why they don't"**



Social retention (and  
monetization)

# KPIs, basics

- Median number of friends
- Correlation Retention/Nber of friends
  - Of course, just saying “people who stayed longer in the game tend to have more friends” is useless, as you can’t have less friends with time
  - So we need a “all things being equal” approach: For example, What is the probability to reach D30 if I have N friends at D7? How this probability evolves if I have P friends with  $P=2*N$
- Ratio Linked friends / All friends
  - Linked friends: the player invited them in the game or was invited by them
- Social actions, total and per friends
  - Nber of time I attacked a friend (compared to random and revenge non-friend), average nber of attack per friend I attacked, percentage of friends I attacked/ I didn’t attack
  - Nber of time I gave, traded, or asked for a sticker, and number of friends involved
  - Specific events (like scary cake) data

# KPIs, basics, even if not social related

- The classic ones (DAU, stickiness, ...) but also:
  - Definition of churn: after how many days without connection would a player statistically leave the game?
  - Nber of days connected per week and Nber of sessions per day per decile
  - This is important to define “timelines” in future features

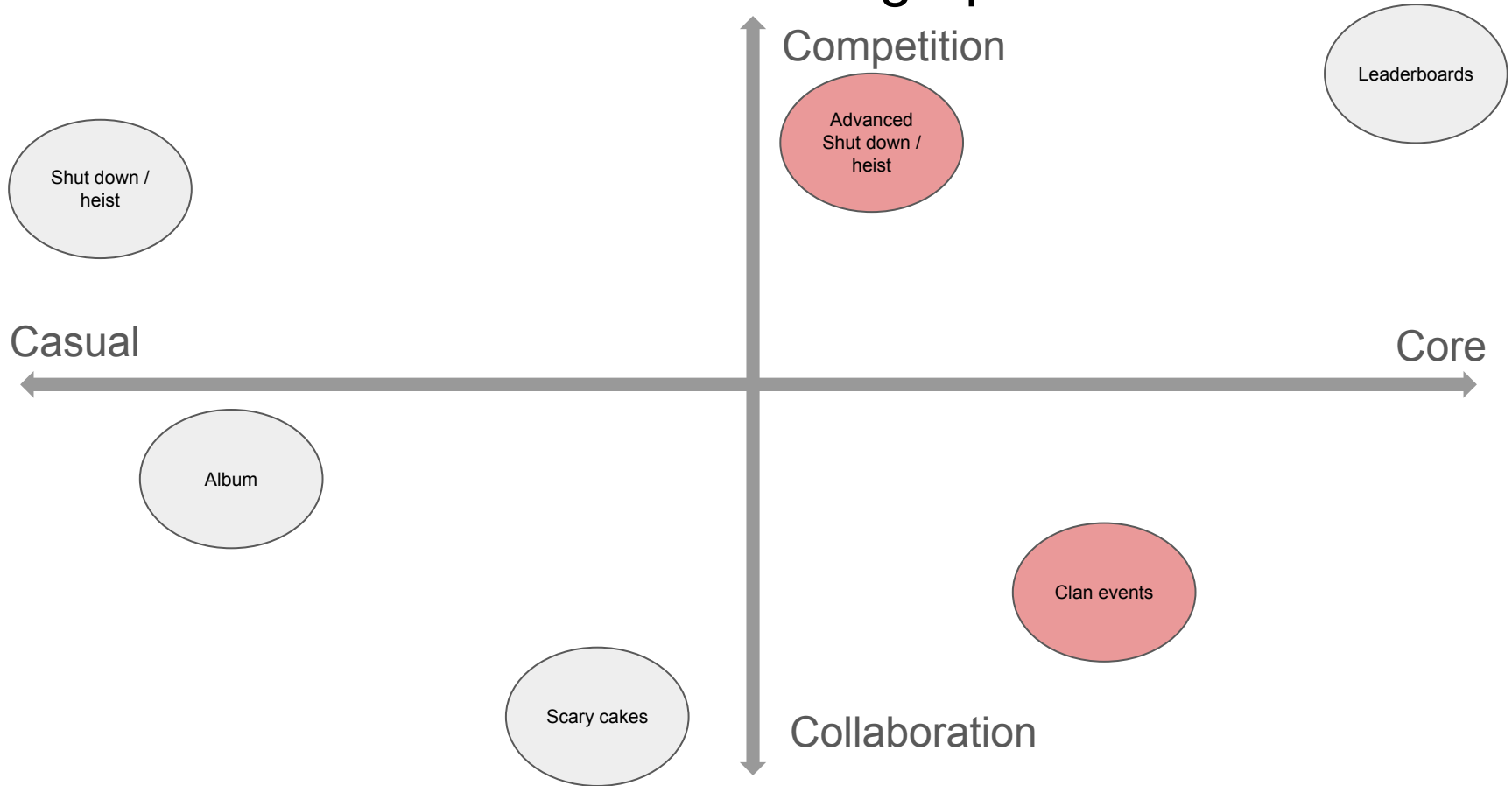
# Demographics

- What are Monopoly Go! Demographics, and which part is covered by social features
- Let's take a very basic typology
  - Achievers: want to beat the game OR collect everything
  - Killers: want to beat other players and brag
  - Explorers: want to discover everything and like new features or new content
  - Strollers: want to have a cool, relaxing walk
  - Socializers: want to interact with others
- Main questions:
  - If achievers are highly represented, are they offered satisfying challenges that request social interactions? Are there “element to collect” only accessible through social interactions?
  - If killers are highly represented, do they choose in-game friends as target? If not, why? Do they have enough opportunities to brag?
  - If explorers are highly represented, do social feature open full new worlds or give better access to existing ones?
  - If strollers are highly represented, do social features make their life easier, more varied, is the walk more enjoyable?
  - And if socializers are highly represented, do we offer enough meaningful social interactions?

# Demographics on a 2 dimensions graph

- If we want an even higher view (and more fluid too), we can try to place our players on a 2 dimensions graph:
  - Do they want more casual actions or more complex actions?
  - Are they looking for more competition or more collaboration?
- Then we can see if the existing social features match or if there is a hole

# Social features on a 2 dimensions graph



# Ideas

# Acquisition: reward

- As we said, “time” is an important factor for social acquisition: time between 2 invitations, time to answer the invitation, and so on
- Dice don’t really have a “time” aspect: what difference if I have these dices in one week or in two weeks?
- Adding cards would add a time value: if i am about to finish a collection, and the collection is about to expire, I will want friends NOW
- Adding event currency from time to time, especially social event currency (candies for scary cake), would have a double effect: the “do it now” effect, like cards, and the opportunity effect: currently, successful invites are more profitable



# Acquisition: new player incentive

- New player incentive highly depends on demographics
- I don't have a strong idea, but examples could be:
  - Protection : for 48h, you can attack but not be attacked (killers)
  - Customization: start the game with a rare token (Monopoly players and achievers)
  - Collection: get 20 marvel sticker packs (Marvel fans and achievers)
  - Benefits: for 7 days, win 10% of your godfather's dice rewards, additional/not subtracted from him

# Why no clan/team/guild?

- You probably have a reason, as they are in other games
- Clan is a good way to create social pressure
- Well handled, social pressure can easily increase number of connections per day and eventually, monetization
  - If there is an easy action one can perform twice a day, and an easy way to check how many of these actions were performed by each clan member these last 7 days, non-parasites clan member will try to reach twice a day
  - Team Packs in which the payer buys a bundle almost at Solo Packs price but can share a small bonus with their Clan are successful in several games and can target different Personas
  - In a very involved clan, if there is a cheap Team Pack, there could even be a pressure to have each member of the Clan buy at least one. And triggering first purchase is often a good way to trigger additional purchases later on
- Clan feature doesn't only bring the benefit of the feature itself but also of features you can build upon it: clan events, clan fights, clan purchases...

# Team Theory

# Basics

This is a sum-up of my own vision, fed with various readings or conference speeches

- To make a Team/Guild meta-feature strong on a global level, several ingredients are needed
- Some exist in games like Coin Master or Royal Match even if not very strong: cooperation, communication...
- But 2 are currently missing: dedication and social management
- Dedication starts when players have to ***make choices*** to help the team
- Social management starts with encouraging good behaviors and punishing bad behaviors

# Dedication

- Level 0: automatic. No dedication.
  - when the game says “build your village to make your team win”, it says “play the game the normal way, and you will help your team”
- Level1: Team task, no cost, no choice. Smallest possible dedication: chooses to do / not to do
  - An example would be: each day, you can click “increase team happiness” for free once. It adds 1% to team happiness until tomorrow. On each money reward, reward is increased by happiness percentage. To get this rewards, the only thing the team players must invest is time
- Level2: Team task, no cost, some choice. Medium dedication: it will not request resources, but some investment in communication may be needed, or some brain energy to figure out the best allocation.
  - An example could be: each day, you can increase a team characteristic for free once, but you have to choose between 2 : happiness and duration. Happiness increases the rewards, duration increases the duration of happiness. Remaining duration is not displayed.
- Level3: Team task with choice between solo and team, with inherent costs/risks . High dedication. The player must choose if they want to serve the greater good (investment) or their own interests.
  - For example: each day you can play some of your dices on the Team Board, which gives you no reward, but increases the Team Level. Team Level determines how much additional money is given daily to Team members

# Constraints

- The need for social management is based on the fact that Team brings more than it costs, but needs everyone efforts to be optimal (Team replaces short term low benefits by mid-term higher benefits)
- To reach this balance, Team features work on constraints. The usual major constraints are “size” and “time”
  - Size: Even in the lowest level of social management, some people are kicked out of teams, because number of places is limited. So having a useless team member has a cost of opportunity
  - Time: usually, what the team wants to achieve is time-framed, either for visible reason (time limited event) or intrinsic reason (being the best team for example requires to progress faster than the other teams).
  - In games like Coin master or Royal match, both size and time are present, but with a low level of pressure
- Additional constraints can be added by the game or by the team-members, and are typically based on investment. Investment has this special ability to open the door for a loss:
  - “If I invest 200 and all the others do, we will all get 300 and i’ll win 100”
  - “if I invest 200 and not all the others do, we will all get 75 and I’ll lose 125. But the parasites will win 75”
- Example of Investment constraints by source
  - added by the game: build team village with your own money
  - added by the other team-members (if it could be checked): each team member must trade or give at least 3 stickers per day, or will be kicked out of the team.

# Social management: punish and reward

- The most basic function to handle social management is “ban”
  - It is very efficient as a punishment
  - It is already a low form of reward (“you have not been excluded this week”)
- Another layer of reward can be added (not necessary but interesting)
  - It’s typically what “roles” do
  - “Roles” allows the team (through its team leader usually) to give special powers and special names to some of the team members. Usually, these powers are delegation of Team Leader powers: ability to recruit, to ban, to launch a team event, to modify team description or requirement, etc.
  - They are not in-game rewards, more “status” reward. Moreover, they give the opportunity for the team to set “example players”
  - Roles are just an example and it’s possible to go beyond that and give more tangible rewards

# Social management: why are parasites good

- Parasites are players that take more from the Team than what they give
- Parasites are *beneficial* to team features:
  - They are the risk, the unknown, one of the reasons why the outcome is uncertain
    - Without uncertainty, there is less interest, and there is definitely less room for “panache”
  - They are *the enemy from inside*, the justification of teammates surveillance/monitoring, for the very existence of punishment tools (ban, ...): banning a teammate for lack of efficiency can be unethical, banning a parasite is just common sense. Parasites bring a *moral value* to social management
  - They are *undefined*: each team has its own definition of where parasitism starts. This is where social pressure takes shape
  - They create room for their opposites, the Champions. In a world of efficiency, top contributors are only over-efficient. In a world where evil parasites exist, they are the True Champions of Good
- That is the reason why any decision to limit parasitism is questionable
  - Saying, as a game, “you must participate that much to access the rewards” is a non-sense for me



# The Mix

- Of course, what makes the strength of all these aspects is their Mix
- It's because players can have different dedications that there can be different rewards (or risks for punishment)
- It's because of constraints that dedication is needed and can vary
- And this Mix creates both extremes: the Champions and the Parasites
- What's the interest for the developer? To replace game's pressure by Team's Social pressure
  - If I need to come everyday to have the reward the game promises me, pressure comes from the game
  - If I need to come everyday to fulfill the goals the Team has set for the event, pressure comes from the Team
- Team social pressure is **stronger and less prejudicial** for the game

# My Idea for Clan

- The closer the friends, the stronger the clan and the social pressure
- A clan also need a “regulation”, a reason to kick players
- So clan should have an evolutive small size: at start, my clan has like 5 slots (including mine), and can evolve until it has 50
- For this and other reasons, Clan also have level
- The level-up is based on an investment VS payback vision: on a daily basis, player must “sacrifice” something (time, dices, money) to contribute to level-up; but depending on the clan level, they also have a daily revenue
  - Example, just for illustration: every day, each clan member can put up to  $Xk$  coins, with  $X$  = their current village number. When there is enough money, the clan levels up. And every day, whatever the village number or the previous investment, each clan member wins “clan level” free dices

# Call to arms

- “Call to arms” are a way to have team events without teams
- Sadly for me, some of “Call to arms” benefits or features already exist in Scary Cake, that’s why I’m so interested in Scary Cake detailed data report
- In “call to arms”, a player can launch a “project” which is too big for him/her
- They can ask their friends to join BUT if the friends do come, they will be able to ask for their own friend, and so on. Player A could be helped by player C who is not from Player’s A friend list, just because Player B answered the call and made player C come.

# Call to arm example with scary cake

- During the event, players can collect candy (same)
- Cake oven are not given. On a specific occasion (example: land on free parking), a cake oven is given. Having your own cake oven prevents you from winning another cake oven (instead, you will earn candies for example)
- Cake oven requires 10 time more points, but you can have up to 19 other players helping you. To find them, you ask your friends. If they join, they will be able to ask theirs, and so on.
- When a reward is unlocked, it's available for the 20 participants, but an additional smaller reward is unlocked only for the oven owner.
- A player can participate in 3 additional “scary cake” at a time, even if focusing on one is more relevant
- Finishing a scary cake releases the corresponding slot, and a scary cake can be abandoned by some of the participants. Their participation stays.
- Cake oven lifespan is lower than event duration. For example, the event would be 7 days, and a cake oven would only last 48h, to create frustration and transfers (I stop working on something that will fail, and focus on a new one that seems promising)

# Why Call to arms

- As I said, Call to arms are a way to have a team feature without team
- Even if less strong, some key elements will appear: champions (that you always invite), parasites (that you don't want in your list), limited size, limited time, so social pressure, etc.
- It's also a way to have an epic friend feature even for players with few friends, as you only need one active friend and a good "chain reaction"

# What to do if we have team? Team Quest

- Team quest events : 5 to 7 goals, based on normal actions (as daily quests), with very high amounts
- Each member of the team can contribute to only 1 goal. They can switch when the goal is finished or even before, but it takes 1h to be “registered” on the new goal (so it’s more efficient to distribute the forces from the beginning) than to do goal 1 with everyone, then goal 2, etc.)

# What to do if we have team? Teams War

- The game chooses the 2 teams (if the event works, we can have a deeper/more complex event with 3 to 5 teams in the same Team War)
- Now, each time you have a “Shut down”, you have a new tab, team war, with members of the opponent team
- Closing a building brings 1 point, hitting a shield brings 0.5 points, and if you don't have a sum of 5 undestroyed levels on your buildings (not each of them, in total), you can't take part.
- Communication in the chat is a way to focus on unshielded opponents
- At the end of the event, the team with more points win. Additional rewards are unlocked by the number of points reached, including for the losing team